ChipletEM: Physics-Based 2.5D and 3D Chiplet Heterogeneous Integration Electromigration Signoff Tool Using Coupled Stress and Thermal Simulation

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ABSTRACT

A review of recent studies on up-to-date IC shows that electromigration (EM) has become one of the major challenges for 2.5D and 3D chiplet heterogeneous integration (CHI) systems. However, most existing researches on EM are focusing on 2D power delivery network without taking Through Silicon Via (TSV) and non-uniformly thermal distribution condition between dies into consideration. To address this problem, this article proposes a novel EM simulation tool ChipletEM for 2.5D and 3D CHI systems. A finite volume method (FVM) based electrical-thermal co-simulation model is employed to get initial temperature and current density inside TSV. And a finite difference time domain (FDTD) solver is used for hydrostatic stress simulation for both nucleation and post-voiding phases. Thermal migration (TM) effect is also considered in the solver. An analytical TSV thermal solver is employed for temperature distribution simulation and thermal dependent current simulation. The FDTD EM solver and TSV thermal solver are coupled together at each time step so that the interaction among EM stress, thermal stress, void growth, resistance change, IR drop and Joule heating effects can be simulated within a single simulation framework. Simulation results show that compared with Finite Element Method (FEM) tool, average error is 0.61% in nucleation phase and 2.4% in growth phase. And the error of proposed method is reduced from 22.22% to 5.24% compared with state of art atomic flux divergence (AFD) method.

KEYWORDS

Chiplet integration systems; TSV; Reliability; Electromigration; Thermal-electrical co-simulation

1 INTRODUCTION

With emergence of high performance AI algorithms, Through Silicon Via (TSV) based 2.5D and 3D chiplet heterogeneous integration (CHI) systems have become one of the most promising technology to achieve the requirement of excessive data communication [1–3]. The architecture of a 2.5D CHI systems is shown in Fig. 1.

However, due to the high current density and non-uniformly distributed high temperature on TSV, a number of reliability issues

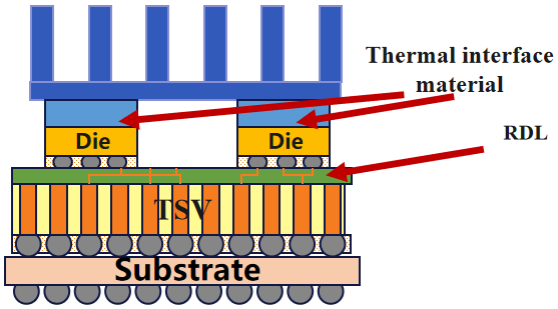


Figure 1: Architecture of a 2.5-D CHI system.

,including electromigration (EM), Joule heating, thermomigration (TM), time-dependent-dielectric-breakdown(TDDB), have become the main threaten to the advanced CHI systems[4–6]. Among these reliability issues, EM is one of the major reliability concerns in the back-end-of-line (BEOL) interconnects, especially in the advanced sub-10nm 2.5D and 3D CHI systems. According to the ITRS 2015 report, EM lifetime will be significantly reduced due to increasing current density and shirking size of net cross sections[7]. Therefore, a fast and accurate EM model is essential for advanced 2.5D and 3D CHI systems. However, existing models cannot meet the requirement of efficiency and accuracy sign-off. Traditional mean-time-to-failure (MTTF) models such as black's equation [8] and blech's product [9] are empirical fitting-based models, which are not scalable and robust enough for the modern 2.5D and 3D CHI systems. The models based on the experiment data of simple one-segment metal line cannot get an accurate EM failure prediction of complicated multi-segment wires and TSVs. Besides, these models are over conservative, bringing unnecessary constraint and increasing the difficulty of IC design.

In order to mitigate the disadvantages of traditional empirical models, a number of physics-based models are proposed [4, 10–14], which considers general interconnect for the 2D interconnect. These models are compact and can solve the stress based partial different equation (PDE) effectively. And the work in [10] has considered the TM and Joule heating effect. However, these methods focus on SoC systems and only consider the Joule heating effect of interconnects and local thermal distribution. As presented by [15–18], global thermal effect is critical and can be one of the dominant factors for EM. In the advanced 2.5D and 3D CHI systems, due to unique

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topology and non-uniformly distributed temperature of TSV, a specific EM analysis tool needs to be developed.

In order to address aforementioned issues, this paper proposes a novel physics-based stress and thermal coupling analysis tool for TSV based 2.5D and 3D ICs. Specifically, the contributions are summarized up as follows:

•A finite difference time domain (FDTD) based EM-TM model, including both nucleation and post-voiding phases, is applied to TSV. Hydro-static stress is calculated by the extended Korhonen's equation combined with both EM and TM flux.

•A global thermal model of CHI system is employed in order to get the accurate temperature distribution. A realistic CHI model ,including heat sink, is implemented. And a finite volume method (FVM) is used for the global electrical-thermal co-simulation for high simulation efficiency while retaining the conservation principles inside thermal conduction equation.

•A coupled EM-TM and Joule heating simulation is employed, so that the simulator can consider the interplay among stress, void growth, resistance change and Joule heating effect in a single simulation framework.

•Simulation result shows that the accuracy of the proposed tool is very high. Compared with commercial finite element method (FEM) based tool, average error is 0.61% in nucleation phase and 2.4% in growth phase. And the error of the proposed method is reduced from 22.22% to 5.24% compared with state of art atomic flux divergence (AFD) method. The error rate is reduced by 76.4%.

•The proposed tool is applied to a 2.5D CHI system. And the simulation result shows that EM issue is more critical when power consumption is high. It can also be observed that in the early stage the dominant factor for EM is temperature while in the late stage the dominant factor is current density.

This paper is organized as follows: Section 2 reviews state of art EM models. Section 3 presents the background of EM. Section 4 introduces the proposed EM thermal co-simulation method. Section 5 demonstrates numerical results obtained from the proposed method. Section 6 concludes this paper.

2 RELATED WORKS

A few EM analysis for TSV based 2.5D and 3D ICs have been proposed recently[4, 13, 19–21]. Specifically, Frank *et al.*[19] and Pak *et al.*[4] put forward AFD based approaches for the TSV. They investigate the mechanism of the EM failure in TSV, showing that the EM void is in the adjacent metal level. Pak's model points out that the resistance change is related to the current density, temperature, cross section area of the TSV and redistribution layers (RDL) interconnect thickness. And a recent AFD based study [13] points out that the EM void is also possible to be nucleated inside the TSV instead of the the junction with RDL interconnects when TSV diameter is small and its aspect ratio is large. Wang's work[20] has considered recovery effect by converting the time variant current into an equivalent square waveform. A response surface methodology based algorithm taking current density, duty cycle and temperature as input parameter is proposed in that work. Nevertheless, the physics mechanisms of EM are not well studied by these methods since internal stress evolution inside the TSV is not counted. Thus the models, not scalable and too conservative, will lead to design redundancy. Besides, as one of the key factors of EM failure, thermal effect and Joule heating need to be taken into consideration. Liu's work[21] studies the relationship between self heating effect and EM of the TSV. The experiment result shows that increased temperature has a positive feedback on the EM effect and will cause thermal run-away effect due to the small cross section area between TSV and RDL. However, this work only focuses on the local Joule heating of TSV lacking the consideration of global thermal effect.

To address the aforementioned issues, a novel coupled EM and thermal simulation tool is proposed in the work, employing Korhonen's equation based physics models to achieve fast and accurate EM lifetime estimation. Also the proposed method has taken global thermal effect of actual complex structure with heat sink of 2.5D and 3D CHI systems into consideration.

3 A REVIEW OF STRESS BASED THREE PHASE EM MODEL

EM is a physical phenomenon of the migration of metal atoms along the direction of the applied electrical field due to momentum exchange between electrons and atoms. During the migration process, hydro-static stress, as well as the stress gradient, generated inside the embedded metal wire due to momentum exchange between lattice atoms will act as a counter-force for atomic migration along the metal line. Before the void nucleated, the stress distribution in the interconnect can be described as follows:

$$\begin{aligned}
&\frac{\partial \sigma}{\partial t} = \frac{\partial}{\partial x}\left[\frac{D_a B\Omega}{k_B T\,(x)}\left(\frac{\partial \sigma}{\partial x} - \frac{eZ\rho j}{\Omega} - \frac{Q}{\Omega T}\frac{\partial T}{\partial x}\right)\right] \\
&BC: \frac{D_a B\Omega}{k_B T\,(x_b)}\left(\left.\frac{\partial \sigma}{\partial x}\right|_{x=x_b} - \frac{eZ\rho j}{\Omega} - \frac{Q}{\Omega T}\frac{\partial T}{\partial x}\right) = 0 \quad 0 < t < t_{nuc} \\
&IC: \sigma\,(x,0) = \sigma_T
\end{aligned} \tag{1}$$

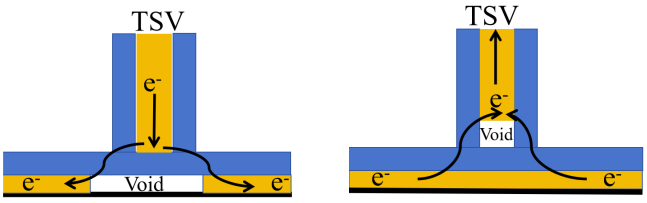
Here, σ is the EM hydrostatic stress, x is the coordinate along the segment length, t is time, $T(x)$ is the temperature at corresponding position, D_a is the effective atomic diffusivity, eZ is the effective charge of migrating atoms, B is the effective bulk elasticity modulus, Ω is the atomic lattice volume, ρ is the copper resistivity, j is the current density, k_B is the Boltzmann constant, T is the temperature, σ_T is initial thermal stress, and x_b is coordinate of boundary node.

When the stress at the cathode exceeds the critical level at t_{nuc}, a void will be nucleated and start growth which is called incubation phase. The normal component of stress at the void boundary is typically zero. Stress gradient between the zero stress void surface and the surrounding is shown as follows:

$$\nabla\sigma(x,t) = \frac{\sigma(x+\delta,t)}{\delta} \qquad on \quad \partial\Omega_L \cap \Gamma_{void} \tag{2}$$

Here δ is effective thickness of the copper-void boundary, Ω_L is the volume of the remaining interconnect wire, Γ_{void} denotes the void boundary, which can be a union of several discrete voids, the $\nabla\sigma(x,t)$ essentially serves as a force to move the void boundary toward the atomic flux direction. The void volume is determined by the stress distribution on the remaining section of the wire and

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(a) Void nucleated in RDL junction (b) Void nucleated inside TSV

Figure 2: Location of EM void in TSV. (2a) is the condition that the void nucleated in the RDL layer and (2b) is the condition that the void nucleated inside TSV

it can be calculated by the following atom conservation equation:

$$V_{void}(t) = -\int_{\Omega_L} \frac{\sigma(t)}{B} dV \tag{3}$$

Here $V_{void}(t)$ is time dependent void volume. When the void volume is larger than critical volume (V_{crit}) which mainly depends on the cross section of the wire, the resistance starts change. That phase is called growth phase.

For the TSV, the void location depends on the directions of electron flow. Based on the experiment result in [13], the location of the void can be nucleated at the junction of TSV and RDL line or inside the TSV with different current directions. Fig. 2a shows the condition that the void nucleated at the junction with the electron flow from TSV to RDL line. The resistance can be calculated with the following equation:

$$\Delta R = \frac{\rho_B}{2 \times \pi \times t_B} \times ln\left(\frac{r_{void}}{r_{TSV}}\right) \tag{4}$$

Here ρ_B is resistivity of the barrier material(Ta/TaN). t_B is thickness of the barrier. r_{void} is radius of void and r_{TSV} is radius of TSV. And Fig. 2b shows the condition that the void nucleated inside TSV with the electron flow from RDL line to TSV. The resistance can be calculated with the following equation:

$$\Delta R = \frac{\rho_B V_{void}}{r_{TSV}^2\left((r_{TSV} + t_B)^2 - r_{TSV}^2\right)} \tag{5}$$

The work focuses on the condition that the void nucleated inside TSV shown in Fig. 2b. And the other conditions will be studied in the future work.

4 METHODOLOGY

4.1 The New Coupled TSV EM Check Flow

In this part, the flow of proposed ChipletEM tool is presented as shown in Fig. 3. The whole flow mainly consists of three major parts: *CHI system electrical-thermal co-simulation model, Fast EM solver* and *TSV thermal solver*. CHI system electrical-thermal co-simulation model is employed in order to get the initial temperature distribution in the CHI systems and current density of each TSV. Fast EM solver is implemented with a FDTD based method. Thermal migration effect is also considered in the model. TSV thermal solver use an analytical model to get temperature distribution and temperature dependent current density inside TSV. Joule heating effect and temperature dependent electrical conductivity are combined together. EM and TSV thermal solver are coupled together. EM induced resistance change will change the Joule heating condition so that the corresponding temperature distribution is also changed. And the time dependent temperature acts as one of the dominant factors in the EM simulation. The time dependent IR drop of CHI systems considering global thermal effect can be demonstrated to the users for further design optimization. The following section will present aforementioned steps in details.

4.2 CHI electrical-thermal co-simulation model

In order to get the thermal distribution and the current density in TSV, an efficient electrical-thermal co-simulation model [18] for CHI systems is employed in this work. Due to the complexity of CHI systems, detailed circuit level based power analysis is not time efficient. Thus, the employed model abstracts the parameters and maps the circuit level thermal model to system level. The Joule heating power is calculated as follows:

$$\begin{aligned} \nabla \cdot [\kappa(x,y,z)\nabla T(x,y,z)] &= -P_{total} \\ P_{Joule} &= \frac{1}{\rho(x,y,z,T)}|\nabla\varphi(x,y,z)|^2 \\ P_{total} &= P_{init} + P_{Joule} \end{aligned} \tag{6}$$

Here κ is thermal conductivity of the solid medium, P is the power consumed by interconnect, x, y, z are the coordinate of CHI system, T is temperature, ρ is the temperature dependent resistivity and φ is the potential distribution, P_{init} is the initial power consumption without Joule heating.

As can be seen, P_{Joule} is affected by temperature dependent resistivity ρ. Meanwhile, P_{Joule} is a part of heat source leading to temperature increasing.

And the temperature dependent resistivity of RDL, bump array, TSV and other interconnects are estimated with the equivalent model to improve the computational efficiency [15–17, 22]. The equivalent model can reduce the impact of complex structure and the composite material of these interconnects and get an estimated resistivity efficiently which is close to the actual resistivity. And the model is discretized by FVM method.

With the presented model, the global thermal effect of CHI system considering convective heat transfer effects and Joule heating effects can be estimated. Individual TSV initial temperature can be obtained from the global thermal distribution.

4.3 Coupled Thermal and Electromigraiton Multi-Physics Simulation

In order to get accurate EM stress for IR-drop analysis, a coupled thermal and EM simulation method is proposed. Temperature is one of the dominant conditions for the EM failure mechanism. It contributes to both stress evolution and TM effect. Meanwhile current induced Joule heating is one of the major heat sources of interconnects. So it is essential to do coupled simulation to get an accurate and EM stress evolution process. In order to improve the accuracy of the EM stress estimation in the full-chip power grid level, we have to consider multi-physics effects and their interaction.

For the Korhonen's equation based EM model from Eq. 1, a finite difference based model is employed for detailed stress analysis over time. Employed method follows the similar work [11, 12, 14]

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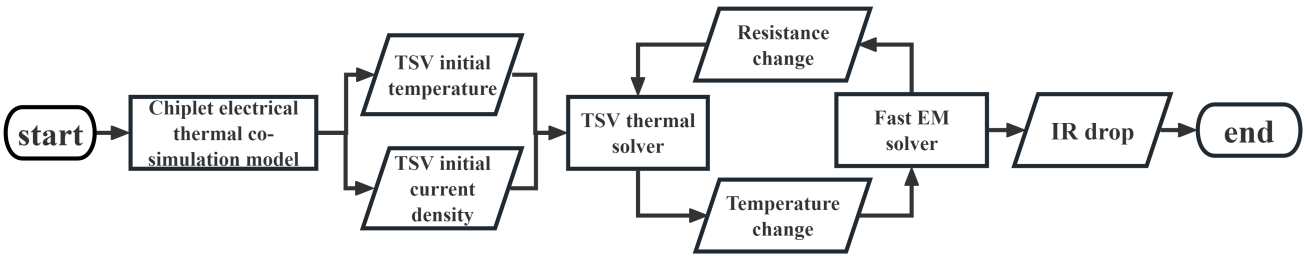


Figure 3: Simulation flow for proposed ChipletEM tool

without the model order reduction feature as shown in the following equation:

$$C\dot{\sigma}(t) = A\sigma(t) + PI(t) - D \tag{7}$$

Here C and A are $n \times n$ matrices. P is the np input matrix. D is $n \times 1$ constant matrix corresponding to TM effect, n is the number of discretization and p is the size of driving current density sources $I(t)$.

For the TSV thermal model, a close form equation from [23] considering Joule heating induced current. And the Joule heating is calculated by the following TSV thermal model in Eq. 8.

The equation is shown as follows:

$$T_{TSV}(x) = \frac{\alpha^2 T_{sub} + \theta}{\lambda^2} + \frac{(\lambda^2 - \alpha^2)T_{sub} - \theta}{\lambda^2(1 + e^{\lambda H})} \times \left[e^{\lambda(H-x)} + e^{\lambda x} \right] \tag{8}$$

Here the T_{sub} is the temperature of substrate, H is height of TSV, x is coordinate of the TSV. And the λ^2, α^2 and θ is presented as follows:

$$\lambda^2 = \frac{1}{k_m}\left[\frac{2k^*_{ins}}{t_{ins} r_{Cu}} - \frac{I^2(x)\rho_m \beta}{\pi^2 r^4_{Cu}}\right]$$

$$\alpha^2 = \frac{2k^*_{ins}}{k_m t_{ins} r_{Cu}}$$

$$\theta = \frac{I^2(x)\rho_m \left[1 - \beta T_{ref}\right]}{k_m \pi^2 r^4_{Cu}}$$

$$I(x) = \frac{V(0) - V(H)}{\int_0^H R_E(x)dx}$$

$$\int_0^H R_E(x)dx = R_0(1 - \beta T_{ref})H + R_0\beta\left[\frac{\alpha^2 T_{sub} + \theta}{\lambda^2} \cdot H + \frac{(\lambda^2 - \alpha^2)T_{sub} - \theta}{\lambda^3(1 + e^{\lambda H})} + \frac{(\lambda^2 - \alpha^2)T_{sub} - \theta}{\lambda^3(1 + e^{\lambda H})} \cdot 2(e^{\lambda H} - 1)\right] + R_{EM} \tag{9}$$

Here $I(x)$ is current, $\int_0^H R_E(x)dx$ is temperature dependent resistance, k_m is thermal conductivity of copper, R_0 is resistance under reference temperature T_{ref}, β is temperature coefficient of copper, ρ_m is resistivity of copper, r_{Cu} is ratio of copper inside TSV, t_{ins} is thickness of insulator, T_{sub} is temperature of substrate, $V(0)$ is the bottom voltage of TSV, $V(H)$ is the top voltage of TSV and R_{EM} is the EM induced resistance.

As can be seen in equation 9, $I(x)$ and $\int_0^H R_E(x)dx$ are coupled together to get the steady state solution of corresponding working condition. And the resistance changed due to EM can be calculated by Eq. 5 and 7. The resistance is updated with the change of EM conditions.

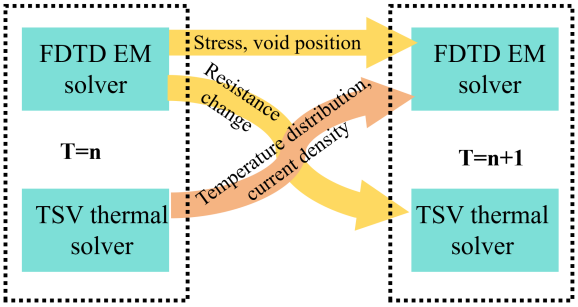


Figure 4: Block diagram of coupled FDTD EM and TSV Joule heating simulator.

As can be seen, Eq. 7, 5 and 9 are coupled together as shown in Fig. 4. Since thermal solver is static, EM simulation time step is used as the coupled simulation time step here. FDTD EM solver passes resistance change to TSV thermal solver and gets temperature distribution and Joule heating induced current density from it. Accurate EM and thermal information can be achieved with the coupled simulation at each time step.

5 NUMERICAL RESULTS AND DISCUSSION

This section presents numerical experiments of the proposed ChipletEM tool which is carried out on a Windows-based simulation platform with an 3.70GHz AMD Ryzen5 5600X processor which has 6 cores. FDTD EM solver and TSV thermal solver are prototyped in MATLAB and C, respectively. The experiment is performed on a 2.5D CHI system published in [18]. This section will first validate the proposed tool with a state-of-art EM model and experiment result; then present global chiplet scale resistance change result.

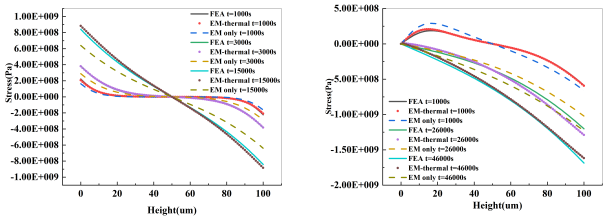
5.1 Accuracy comparison against existing EM analysis methods and experiment results on single TSV

This part first compares the result of the proposed method and FEA tool, then compares the proposed method with state-of-art EM models and experiment result.

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Table 1: Failure rate of high and low power mode

	Nucleation phase	Growth phase
EM-thermal average error	0.61%	2.4%
EM-thermal max error	2.62%	6.67%
EM only average error	3.48%	10.03%
EM only max error	12%	26.79%
TM contribution	11.4%	15.83%



(a) Nucleation phase (b) Growth phase

Figure 5: Coupled EM-thermal simulation result

5.1.1 Validation with FEM tool. Fig. 5 shows the comparison result between FEA tool, EM-thermal co-simulation method and EM only simulation method in nucleation phase and growth phase. Experiment is performed on a single TSV with 100*um* height. As can be seen in Table 1, the average error is 0.61% in nucleation and 2.4% in growth phase. And if only EM is considered without coupled thermal simulator, average error is 3.48% for nucleation phase and 10.43% for growth phase. Max error can reduce from 26.79% to 6.67% with the coupled EM-thermal simulation. It draws a conclusion that Joule heating is one of the key factor for EM and EM-thermal co-simulation can significantly improve the accuracy.

Besides, Table 1 also shows the stress contribution of TM effect. Simulation result shows that TM effect contributes 11.4% in the nucleation phase stress evolution and 15.83% in growth phase stress evolution. And this agrees with the conclusion from experiment in [13] saying that TM effect is relatively weak.

5.1.2 Validation with experiment data. This part validates the simulation result with published experiment data from [13]. Based on the experiment result, $\frac{\Delta R}{R}$ reaches 20% at 32.4 hours with $200°C$ and $1.5MA/cm^2$ current density. Fig. 6 compares the simulation result of the proposed method and AFD method from [13] with real EM experiment result.

It can be seen that in the experiment the TSV reaches 20% resistance change in 32.5 hours. And the simulation result of proposed method is 30.7 hours and AFD method is 25.2 hours. The error of proposed method is 5.24% and the error of AFD method is 22.22%. The error rate is reduced by 76.4%.

5.2 Simulation results on the CHI System

This part demonstrates the simulation results on a 2.5D CHI system which has nine chiplets. The parameters of the system is shown in Table 2 and its layout of the system is in Fig. 7.

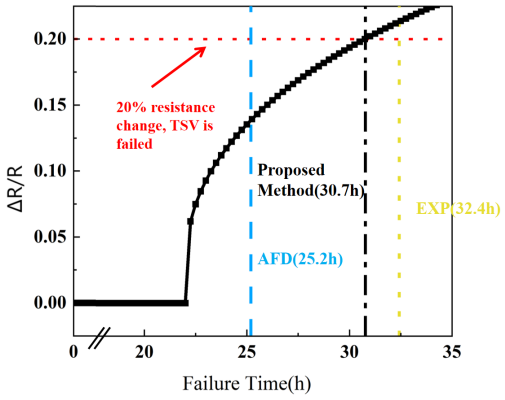


Figure 6: Resistance change trend of TSV

Table 2: Parameters of the CHI system

Name	Number	Size (mm)
Chiplet	9	$8 \times 8 \times 0.15$
Heat Sink	1	$40 \times 40 \times 0.18$
TSV	1600	Diameter = 0.01 Height = 0.1
MicroBump	2500	/

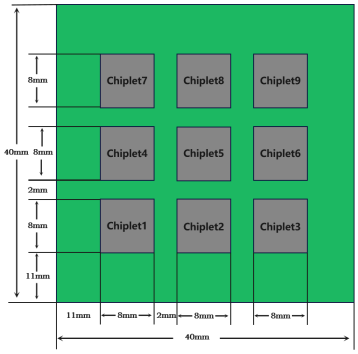


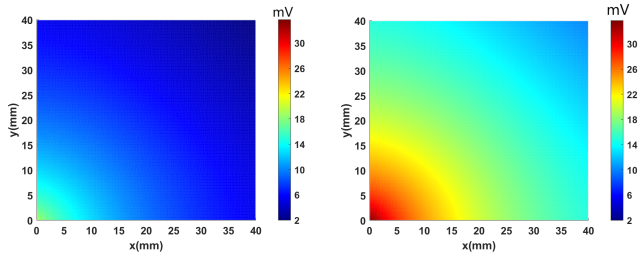
Figure 7: Layout of CHI system

There are low and high working modes for the CHI system. The voltage distribution of both modes are shown in Fig. 8. And temperature distribution results from CHI system electrical-thermal co-simulation model is shown in Fig. 9. It can be seen that the left corner of the chip has high voltage and the medium of the chip has high temperature.

The resistance change distribution of the CHI system of low and high power modes is shown in Fig. 10 and 11, in which the EM issue in the high power mode is more critical than that in the low power mode. And we can see current density and thermal are both critical issues for the EM. It can be observed that in the early stage the high temperature part has more critical EM issue. In the late stage EM issue is more critical in the high current density part.

Table 3 displays that the percentage of failed TSVs is increasing over time. In the low power mode, there is no TSV failed in the first year since most TSVs are still in nucleation phase. And 47%

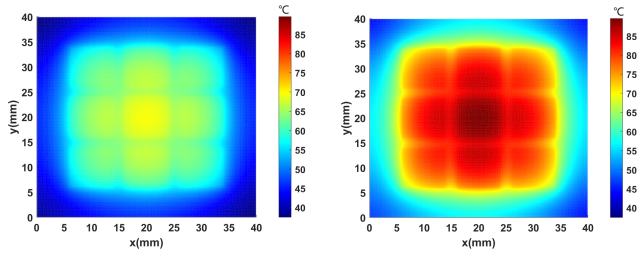
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(a) Voltage distribution for low power mode

(b) Voltage distribution for high power mode

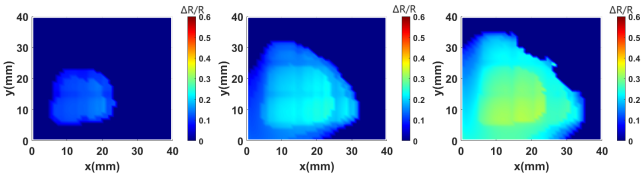
Figure 8: Voltage distribution of CHI system



(a) Temperature distribution for low power mode

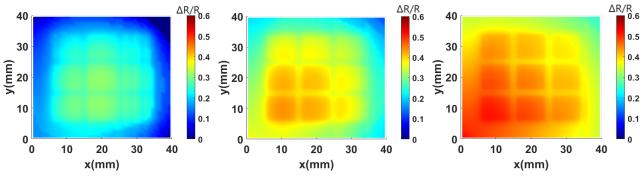
(b) Temperature distribution for high power mode

Figure 9: Temperature distribution of CHI system



(a) 1 year (b) 3 years (c) 6 years

Figure 10: Resistance change distribution of low power mode in 1, 3 and 6 years



(a) 1 year (b) 3 years (c) 6 years

Figure 11: Resistance change distribution of high power mode in 1, 3 and 6 years

TSVs failed in 6 years. EM issue is more critical in high power mode. 54.25% TSVs failed in first year and all the TSVs failed in 6 years.

Table 3: Failure rate of high and low power mode

	6 month	1 year	3 years	6 years
Low power	0%	0%	16.75%	47%
High power	26.19%	54.25%	91.5%	100%

6 CONCLUSION

This paper has proposed a novel EM simulation tool ChipletEM for 2.5D and 3D CHI systems. A FVM based electrical-thermal co-simulation model is employed to get initial temperature and current density inside TSV. A FDTD EM solver and a compact TSV thermal solver is employed for coupled simulation for interaction among EM stress, thermal stress, void growth, resistance change, IR drop and Joule heating effects. Accuracy of the proposed tool is validated with FEM tool and published experiment data. Compared with state of art AFD method the error rate is reduced by 76.4%. In the future the EM failure between RDL and TSV junction will also be taken into consideration.

REFERENCES

[1] Z. Wang, J. Sun, A. Goksoy, S. K. Mandal, Y. Liu, J.-S. Seo, C. Chakrabarti, U. Y. Ogras, V. Chhabria, J. Zhang, and Y. Cao, "Exploiting 2.5d/3d heterogeneous integration for ai computing," in *2024 29th Asia and South Pacific Design Automation Conference (ASP-DAC)*, pp. 758–764, 2024.

[2] G. Krishnan, S. K. Mandal, M. Pannala, C. Chakrabarti, J.-S. Seo, U. Y. Ogras, and Y. Cao, "Siam: Chiplet-based scalable in-memory acceleration with mesh for deep neural networks," *ACM Trans. Embed. Comput. Syst.*, vol. 20, sep 2021.

[3] Y. S. Shao, J. Clemons, R. Venkatesan, B. Zimmer, M. Fojtik, N. Jiang, B. Keller, A. Klinefelter, N. Pinckney, P. Raina, S. G. Tell, Y. Zhang, W. J. Dally, J. Emer, C. T. Gray, B. Khailany, and S. W. Keckler, "Simba: Scaling deep-learning inference with multi-chip-module-based architecture," in *Proceedings of the 52nd Annual IEEE/ACM International Symposium on Microarchitecture*, MICRO '52, (New York, NY, USA), p. 14–27, Association for Computing Machinery, 2019.

[4] J. Pak, S. K. Lim, and D. Z. Pan, "Electromigration study for multiscale power/ground vias in tsv-based 3-d ics," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 33, no. 12, pp. 1873–1885, 2014.

[5] A. Jain, R. E. Jones, R. Chatterjee, and S. Pozder, "Analytical and numerical modeling of the thermal performance of three-dimensional integrated circuits," *IEEE Transactions on Components and Packaging Technologies*, vol. 33, no. 1, pp. 56–63, 2010.

[6] H.-J. Choi, S.-M. Choi, M.-S. Yeo, S.-D. Cho, D.-C. Baek, and J. Park, "An experimental study on the tsv reliability: Electromigration (em) and time dependant dielectric breakdown (tddb)," in *2012 IEEE International Interconnect Technology Conference*, pp. 1–3, 2012.

[7] ITRS, "International technology roadmap for semiconductors (itrs)," Mar. 2015.

[8] J. Black, "Electromigration—a brief survey and some recent results," *IEEE Transactions on Electron Devices*, vol. 16, no. 4, pp. 338–347, 1969.

[9] I. A. Blech, "Electromigration in thin aluminum films on titanium nitride," *Journal of Applied Physics*, vol. 47, pp. 1203–1208, 04 1976.

[10] L. Chen, S. X.-D. Tan, Z. Sun, S. Peng, M. Tang, and J. Mao, "A fast semi-analytic approach for combined electromigration and thermomigration analysis for general multisegment interconnects," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 40, no. 2, pp. 350–363, 2021.

[11] Z. Sun, S. Yu, H. Zhou, Y. Liu, and S. X.-D. Tan, "Emspice: Physics-based electromigration check using coupled electronic and stress simulation," *IEEE Transactions on Device and Materials Reliability*, vol. 20, no. 2, pp. 376–389, 2020.

[12] C. Cook, Z. Sun, E. Demircan, M. D. Shroff, and S. X.-D. Tan, "Fast electromigration stress evolution analysis for interconnect trees using krylov subspace method," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 26, no. 5, pp. 969–980, 2018.

[13] Y. Zhang, W. Tian, H. Wang, X. Zhang, W. Shao, B. Zhou, Y. Shi, W. Gong, R. Deng, J. Zhao, P. Wang, and Y. Feng, "Electrothermal reliability analysis of electromigration in 3-d tsv-rdl interconnects," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 14, no. 1, pp. 157–165, 2024.

[14] S. Lamichhane, M. Kavousi, and S. X.-D. Tan, "Emspice 2.0: Multiphysics electromigration analysis tool for beyond moore ics," in *2024 IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, pp. 571–576, 2024.

[15] C. Nie, Q. Xu, and L. Chen, "Equivalent thermal model of through silicon via and bump for advanced packaging of integrated circuits," *Microelectronics and reliability*, vol. 137, no. Oct., pp. 178–189, 2022.

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[16] C. Nie, Q. Xu, C. Wang, H. Cao, J. Liu, and Z. Li, "Efficient transient thermal analysis of chiplet heterogeneous integration," *Applied thermal engineering: Design, processes, equipment, economics*, 2023.
[17] C. Wang, Q. Xu, C. Nie, H. Cao, J. Liu, and Z. Li, "An efficient thermal model of chiplet heterogeneous integration system for steady-state temperature prediction," *Microelectronics Reliability*, 2023.
[18] X. Ma, Q. Xu, C. Wang, H. Cao, J. Liu, D. Zhang, and Z. Li, "An electrical–thermal co-simulation model of chiplet heterogeneous integration systems," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, pp. 1–13, 2024.
[19] T. Frank, S. Moreau, C. Chappaz, L. Arnaud, P. Leduc, A. Thuaire, and L. Anghel, "Electromigration behavior of 3d-ic tsv interconnects," in *2012 IEEE 62nd Electronic Components and Technology Conference*, pp. 326–330, 2012.
[20] S. Wang, T. Kim, Z. Sun, S. X.-D. Tan, and M. B. Tahoori, "Recovery-aware proactive tsv repair for electromigration lifetime enhancement in 3-d ics," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 26, no. 3, pp. 531–543, 2018.
[21] Y. Liu, M. Li, M. Jiang, D. W. Kim, S. Gu, and K. N. Tu, "Joule heating enhanced electromigration failure in redistribution layer in 2.5d ic," in *2016 IEEE 66th Electronic Components and Technology Conference (ECTC)*, pp. 1359–1363, 2016.
[22] C. Wang, Q. Xu, C. Nie, H. Cao, J. Liu, D. Zhang, and Z. Li, "A multiscale anisotropic thermal model of chiplet heterogeneous integration system," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 32, no. 1, pp. 178–189, 2024.
[23] J. Chai, G. Dong, Z. Mei, and W. Zhu, "Steady state electrical–thermal coupling analysis of tsv," *Journal of Semiconductors*, 2018.

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